The due date for submitting this assignment has passed.
Due on 2026-04-29, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-29, 11:51 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1) Consider the following function **1 point**

$$f(x, y, z) = x^2z + y^3e^{xy} + \sin(xyz).$$

Which of the following option(s) is (are) true?

- ( ) $f_{xx} = 2z + y^5e^{xy} - yz\sin(xyz)$
- (•) $f_{xx} = 2z + y^5e^{xy} - (yz)^2\sin(xyz)$
- ( ) $f_{zy} = \cos(xyz)x^2yz$
- ( ) $f_{xz} = 2x - xy^2z\sin(xyz)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$f_{xx} = 2z + y^5e^{xy} - (yz)^2\sin(xyz)$

2) Number of critical points of the function $f(x, y, z) = e^x(x^2 - y^2 - 2z^2)$ is

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

3) Choose the correct statement(s) about $f(x, y) = \sqrt{x^2 + y^2}$. **1 point**

- [ ] f has no critical points.
- [x] $(0, 0)$ is the only critical point of f.
- [ ] $(0, 0)$ is a local maximum of f.
- [x] $(0, 0)$ is a local minimum of f.
- [x]

$(0, 0)$ is a global minimum of f.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$(0, 0)$ is the only critical point of f.
$(0, 0)$ is a local minimum of f.
$(0, 0)$ is a global minimum of f.

4) Find the minimum value of the surface area of a rectangular box that has a volume of 1000 unit^3.

Note: If x, y and z represent the lengths of the three sides of a rectangular box, then the surface area and volume are given by $2(xy + yz + zx)$ and xyz, respectively.

600

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 600

1 point

5) Consider a function $f(x, y, z) = x^3 + 3y^3 + z^3 - 4x - 6y - 4z$. If A is the the Hessian matrix of $f(x, y, z)$ at (0,1,1), then which of the following option(s) is(are) true?

1 point

- [x] $det(A) = 0$
- [x] $Rank(A) = 2$
- [x] $Nullity(A) = 1$
- [ ] $det(A) = 648$
- [ ] $Rank(A) = 3$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$det(A) = 0$
$Rank(A) = 2$
$Nullity(A) = 1$

6) Consider the following function **1 point**

$$f(x, y) = e^x + e^y - e^{x+y}$$

Which of the following option(s) is (are) true?

- [ ] Hessian matrix of f at point $(1, 1)$ is $\begin{bmatrix} 1-e & -e^2 \\ -e^2 & 1-e \end{bmatrix}$.
- [x] Hessian matrix of f at point $(1, 1)$ is $\begin{bmatrix} e-e^2 & -e^2 \\ -e^2 & e-e^2 \end{bmatrix}$.
- [ ] f has a point of maximum.
- [x] f has a saddle point.

Yes, the answer is correct.
Score: 1

Accepted Answers:

Hessian matrix of f at point $(1, 1)$ is $\begin{bmatrix} e-e^2 & -e^2 \\ -e^2 & e-e^2 \end{bmatrix}$.

f has a saddle point.

7) Choose the correct statements. **1 point**

- [ ] If $f : \mathbb{R}^3 \to \mathbb{R}$ then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in \mathbb{R}^3$.
- [x] If $f : \mathbb{R}^3 \to \mathbb{R}$ is a linear transformation then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in \mathbb{R}^3$.
- [x] For each $n \in \mathbb{N}$, there exists a map $f : \mathbb{R}^n \to \mathbb{R}$ such that the rank of the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is n for some $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.
- [x]

For each $n \in \mathbb{N}$, there exists a map $f : \mathbb{R}^n \to \mathbb{R}$ such that the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is symmetric for all $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If $f : \mathbb{R}^3 \to \mathbb{R}$ is a linear transformation then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in \mathbb{R}^3$.

For each $n \in \mathbb{N}$, there exists a map $f : \mathbb{R}^n \to \mathbb{R}$ such that the rank of the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is n for some $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

For each $n \in \mathbb{N}$, there exists a map $f : \mathbb{R}^n \to \mathbb{R}$ such that the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is symmetric for all $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

8) The maximum value of the function $-2x^2 + 2xy - y^2 + 12x - 4y - 7$ is

13

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 13

1 point

9) Define a function

1 point

$$f(x, y) = \begin{cases} (x-y)^2 \sin\left(\frac{1}{x-y}\right) & \text{if } x = y \\ 0 & \text{if } x = y. \end{cases}$$

Choose the set of correct options.

- [x] f is continuous at $(0, 0)$.
- [ ] $f_x(0, 0) = f_y(0, 0)$
- [x] $f_x(0, 0) = f_y(0, 0)$
- [x] f is differentiable at $(0, 0)$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

f is continuous at $(0, 0)$.
$f_x(0, 0) = f_y(0, 0)$
f is differentiable at $(0, 0)$.

10) A company produces two types of chocolates, one that sells for Rs.3 and another at Rs.2. The total revenue, in lakhs, from the sale of x lakhs chocolates at Rs.2 each and y lakhs at Rs. 3 each is given by

$$F(x, y) = 2x + 3y.$$

The total cost in lakhs for producing x lakhs of the Rs.2 chocolates and y lakhs of the Rs.3 chocolates is given by

$$G(x, y) = 2x^2 - 2xy + y^2 - 9x + 6y + 7.$$

Then the maximum profit in lakhs is

[Enter your answer upto two decimal places]
11.25

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 11.25

1 point

11) (MSQ) Let $f : \mathrm{R}^2 \to \mathrm{R}$ be a multivariable function and $(x_0, y_0) \in \mathrm{R}^2$. Consider the following statements: **1 point**

P: f is continuous at (x_0, y_0).
Q: The partial derivatives of f exist and are continuous in a neighborhood of (x_0, y_0).
R: The tangent plane to the graph of f exists at (x_0, y_0).
Choose the correct statements from the following.

- [ ] P implies that f is differentiable at (x_0, y_0).
- [x] If f is differentiable at (x_0, y_0), then P is satisfied.
- [x] Q implies that f is differentiable at (x_0, y_0).
- [ ] If f is differentiable at (x_0, y_0), then Q is satisfied.
- [x] R implies that f is differentiable at (x_0, y_0).
- [x] If f is differentiable at (x_0, y_0), then R is satisfied.

Yes, the answer is correct.
Score: 1

Accepted Answers:

If f is differentiable at (x_0, y_0), then P is satisfied.
Q implies that f is differentiable at (x_0, y_0).
R implies that f is differentiable at (x_0, y_0).
If f is differentiable at (x_0, y_0), then R is satisfied.

12) (NAT) A particle moving in a potential field tends to settle at points where the potential energy is a local minimum (stable equilibrium point). Consider a particle moving in $\mathbb{R}^2$ where the potential energy as a function of position is given by

$$V(x, y) = x^4 + y^4 - 4xy + 1,$$

for all $(x, y) \in \mathbb{R}^2$. Using the Hessian test, find the potential energy at stable equilibrium points.

-1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -1

1 point